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Euclidean Geometry In Mathematical Olympiads

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Problem 1.40 (Canada 1991/3)



Chapter 1

fgm

fgmcon



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277546

1607 posts

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#1

Problem 1.40 (1991/3) Let P be a point inside circle w . Consider the set of chords w that contain P . Prove That their midpoint all lie on a circle.

277546

1607 posts

Mar 9, 2018, 11:34 AM • 5 👍

#2

Solution



Let the midpoint of wP be called O . Let a generic chord be called AB . The midpoint of AB is to be called M . As AB is a chord of the circle, wM is the perpendicular bisector of AB . Thus, we see that $OM = wM$ and the result follows

enhanced

509 posts

Mar 9, 2018, 12:48 PM • 6 👍

#3

SOLUTION



let AB be a arbitrary chord through P let M be its midpoint so $\angle OMP = 90^\circ, \implies M$ lies on a circle with diameter OP as chord varies ■

WizardMath

2487 posts

Mar 10, 2018, 1:56 AM • 10 👍

#4

solution



The image of the circle w under the homothety at P and ratio $\frac{1}{2}$ is another circle so done.

Remark: ω is better than w in my opinion.

Kagebaka

3001 posts

Sep 30, 2018, 4:31 PM • 2 👍

#5

Can someone explain why @above's proof works?

Maths_1729

390 posts

Mar 6, 2020, 9:23 AM • 1 👍

#6



realquarterb wrote:

Problem 1.40 (1991/3) Let P be a point inside circle w . Consider the set of chords w that contain P . Prove That their midpoint all lie on a circle.

Let $M_1, M_2, M_3, \dots, M_n$ be the midpoints of all the chords passing through the point P and O be the centre of circle then as

$$\angle OM_n P = 90$$

for every n hence as

$$\angle OM_1 P = \angle OM_2 P = \dots = \angle OM_n P = 90$$

which the angle made by same segment AP hence all the midpoints lie on a circle with

diameter AP

mathlogician

1050 posts

Mar 18, 2020, 7:52 PM • 1 👍

#7

solution



We use the same definitions as in Post #2. Let M be the midpoint of chord AB . Clearly, Mw is perpendicular to AB , so $\angle AMw = 90$. Consider the circle with center O and radius OP . Since Pw is a diameter, any point K such that $\angle PKw = 90$ is on the circle, so M is on the circle.

Can someone check this please?

This post has been edited 1 time. Last edited by mathlogician, Mar 18, 2020, 7:52 PM

Kuroshio

72 posts

Mar 22, 2020, 3:08 PM • 1 👍

#8

Consider the circle with diameter OP let's say ω .

Let AB be any chord passing through P with centre M